SALES PERFORMANCE ANALYTICS

Findings and Business Recommendations

Dataset rows	9,994 records
Distinct orders	5,009
Analysis period	Jan 2011 - Dec 2014
Dashboard focus	All categories and all regions (full dataset)
Tools	Python, MySQL and Microsoft Excel

Scope note: The final dashboard is unfiltered, so it shows every category and every region. That means the dashboard totals and the full-dataset totals in this report are the same numbers, and every figure here was checked against the cleaned_sales_data sheet. Profit figures come straight from the Profit column in that sheet.

1. Objective and Dataset

This report goes through four years of sales and tries to answer three plain questions: where is the business growing, where is it making money, and where is money quietly leaking out? The workbook holds 9,994 transaction lines with order and ship dates, customer and location details, product category, sales, quantity, discount and profit. The records run from January 2011 to December 2014.

Over that period the business recorded **2,297,200.86** in sales and **286,397.02** in profit (a 12.47% margin) from **5,009** distinct orders placed by 793 customers.

The data looks like the widely used Superstore sample dataset, but the workbook itself does not say where it came from. This report therefore does not call it real or synthetic. The source should be stated clearly in the project README so nobody mistakes it for a live business.

2. Data Preparation and Validation Summary

Before analysing anything, I ran a set of checks on the cleaned_sales_data sheet and on the summary tables behind the dashboard:

- No missing values in any of the 22 columns, no duplicate rows, and every Row ID is unique.
- Order dates run from 04 Jan 2011 to 31 Dec 2014, and the Month column matches the order date on every row.
- The dashboard KPIs (sales 2,297,200.86, 5,009 orders, average order value 458.61) match the cleaned data exactly.
- The month, region and category summary tables each add up to the same 2,297,200.86.
- The TOP_10 DM sheet was compared with a ranking of all 1,841 product names. It does contain the true top 10, although its rows are listed alphabetically, so this report re-ranks them by sales.

Scope check: Earlier dashboard versions were filtered to Technology only. The current dashboard has no filter applied, so the headline numbers and the category, region and month tables all describe the whole business.

3. Key Performance Indicators

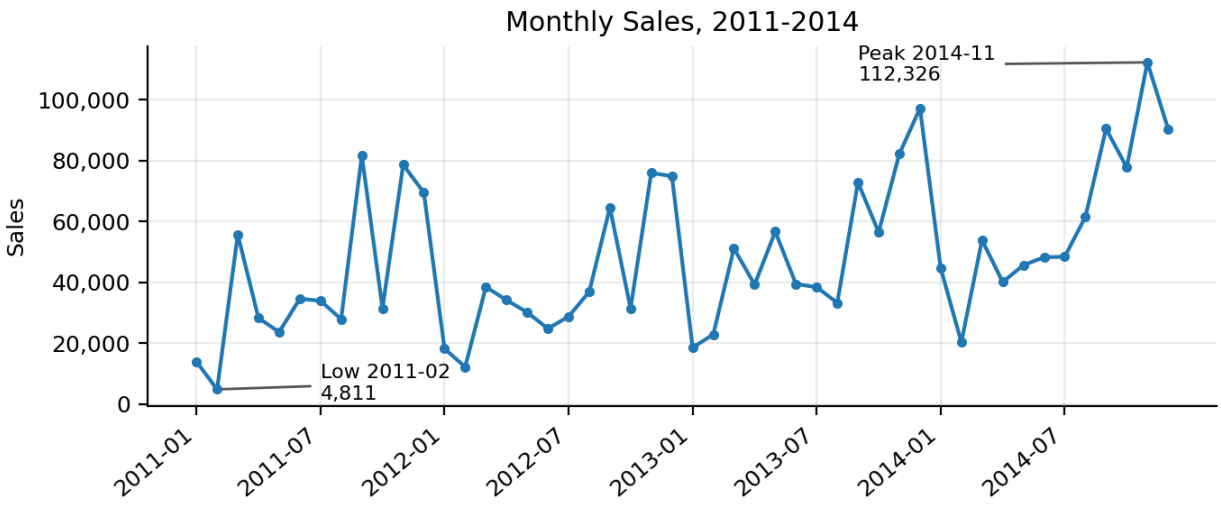
Metric	Value
Total sales	2,297,200.86
Total profit	286,397.02
Profit margin	12.47%
Distinct orders	5,009
Average order value	458.61
Units sold	37,873
Customers	793
Average discount per order line	15.62%

Average order value is sales divided by distinct orders. Profit margin is profit divided by sales.

4. Key Findings

4.1 Monthly Sales and Yearly Growth

Sales were not a straight line upward. They slipped 2.8% in 2012, then grew 29.3% in 2013 and 20.6% in 2014. By 2014 yearly sales reached 733,947.02, which is 51.6% above 2011.



The weakest month was **2011-02** at 4,810.56 and the strongest was **2014-11** at 112,326.47. A typical month brought in about 47,858.35. There is a clear seasonal shape: September, November and December together account for 43.1% of all sales, and the final quarter alone brings 38.2%. February is the quietest calendar month at 2.6% of sales.

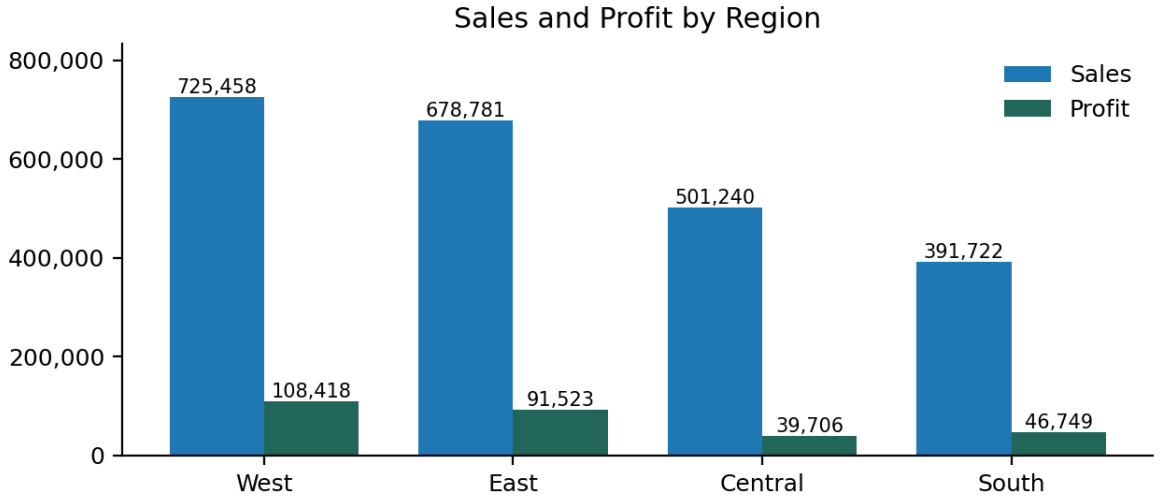
Year	Sales	Growth vs prior year	Orders	Avg order value	Profit	Margin
2011	484,247.50	-	969	499.74	49,543.97	10.23%
2012	470,532.51	-2.8%	1,038	453.31	61,618.60	13.10%
2013	608,473.83	+29.3%	1,310	464.48	81,726.93	13.43%
2014	733,947.02	+20.6%	1,692	433.77	93,507.51	12.74%

The growth came from **more orders, not bigger ones**. Orders rose from 969 in 2011 to 1,692 in 2014 (up 74.6%), while average order value fell from 499.74 to 433.77 (down 13.2%). Margin improved from 10.23% in 2011 to a high of 13.43% in 2013, then eased to 12.74% in 2014.

Interpretation: the seasonal pattern is useful for planning, but the workbook cannot tell us why the peaks and dips happen. Promotions, holidays, product launches and stock levels would be needed to explain them.

4.2 Regional Performance

West is the largest region with 725,457.82 in sales (31.58% of the total), followed by **East** at 678,781.24 (29.55%). **Central** brings in 501,239.89 (21.82%) and **South** is the smallest at 391,721.91 (17.05%). West and East together make up 61.13% of sales.



Region	Sales	Share	Profit	Margin	Orders	Avg order value	Avg discount
West	725,457.82	31.58%	108,418.45	14.94%	1,611	450.32	10.93%
East	678,781.24	29.55%	91,522.78	13.48%	1,401	484.50	14.54%
Central	501,239.89	21.82%	39,706.36	7.92%	1,175	426.59	24.04%
South	391,721.91	17.05%	46,749.43	11.93%	822	476.55	14.73%

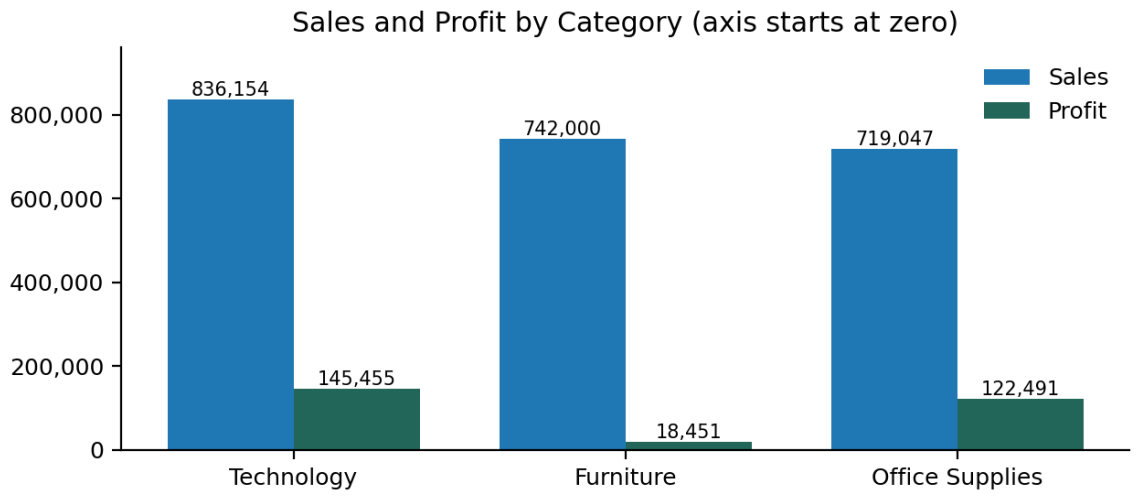
The ranking changes once profit enters the picture. Central is third in sales but earns only 39,706.36 in profit, a margin of **7.92%**, compared with 14.94% in West. Central also has by far the highest average discount (24.04%, against 10.93% in West). That does not prove discounting is the cause, but the two move together and Section 4.5 looks at it more closely.

Central has also stalled. Its sales were flat in 2014 (-0.2%) after a jump in 2013, while West grew 34.0%, South 31.5% and East 18.1%. South had a rough 2012 (-31.3%) but has recovered strongly since. Its margin of 11.93% is respectable, so its main gap is volume rather than profitability.

Revenue alone does not tell us whether these differences come from order volume, customer mix, product mix or pricing. The recommendations below suggest how to dig into that.

4.3 Category Performance

Technology is the largest category at 836,154.03 (36.40% of sales), then **Furniture** at 741,999.80 (32.30%) and **Office Supplies** at 719,047.03 (31.30%). The three are closer than the dashboard chart suggests: the category chart on the dashboard starts its axis at 650,000, which makes Technology look more than twice the height of Office Supplies when it is really only 16.3% larger.



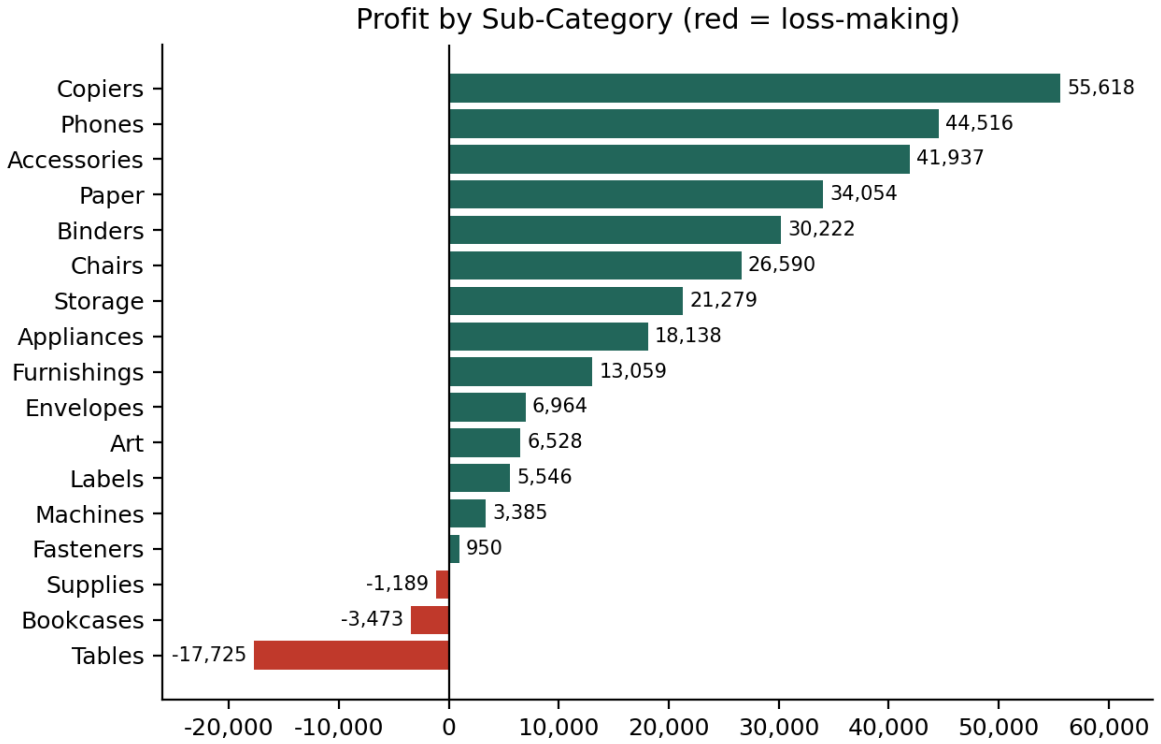
Category	Sales	Share	Profit	Margin	Share of total profit	Orders*	Avg order value*
Technology	836,154.03	36.40%	145,454.95	17.40%	50.79%	1,544	541.55
Furniture	741,999.80	32.30%	18,451.27	2.49%	6.44%	1,764	420.63
Office Supplies	719,047.03	31.30%	122,490.80	17.04%	42.77%	3,742	192.16

*One order can contain items from more than one category, so category order counts overlap and add up to more than 5,009.

The standout is **Furniture**. It delivers 32.30% of sales but only 6.4% of profit, with a margin of just 2.49%. Technology and Office Supplies both earn close to 17% margins. Office Supplies works differently from the other two: it has the most orders (3,742) but small baskets (average 192.16), whereas Technology orders average 541.55.

4.4 Sub-Category Profitability

Going one level deeper explains most of the category story. Three sub-categories lose money overall: **Tables** (-17,725.48, margin -8.56%), **Bookcases** (-3,472.56) and **Supplies** (-1,189.10). Tables is the big one: it sells 206,965.53 yet loses money, with an average discount of 26.13%. By region, Tables lose money in Central, East and South, and only just break into profit in West.



Machines deserve attention too. They sell 189,238.63 but make only 3,384.76 in profit (a 1.79% margin) with a 30.61% average discount. On the bright side, **Copiers** earn 55,617.82 at a 37.20% margin, **Accessories** earn 41,936.64 at 25.05%, and **Paper** earns 34,053.57 at 43.39%. **Phones** is the biggest sub-category by sales (330,007.05) and also a solid earner (44,515.73).

4.5 Discounts and Profit

This is the clearest pattern in the whole dataset. Order lines sold with no discount (4,798 of them) earn a margin of **29.51%**. As the discount grows, the margin falls away and then turns negative.

Discount band	Order lines	Sales	Profit	Margin	Loss-making lines
No discount	4,798	1,087,908.47	320,987.60	29.51%	0
Up to 20%	3,803	846,522.24	100,785.47	11.91%	523
21-40%	460	234,137.90	-35,817.47	-15.30%	415
41-60%	215	71,048.21	-28,944.19	-40.74%	215
61-80%	718	57,584.04	-70,614.40	-122.63%	718

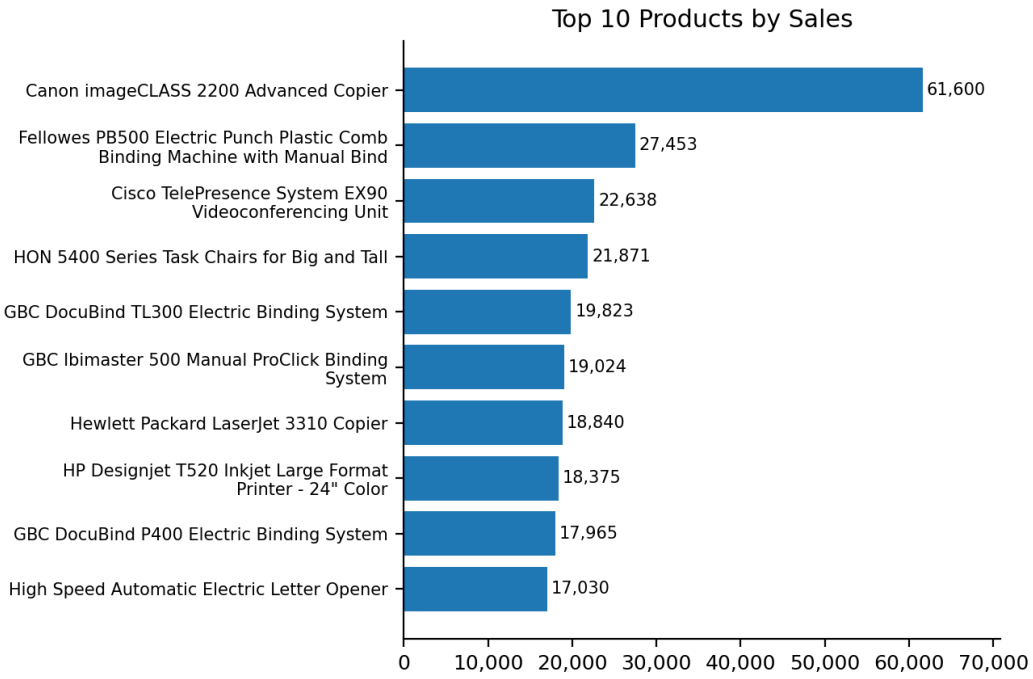
Every one of the 933 lines discounted by more than 40% lost money, 99,558.59 in total on 128,632.25 of sales. Across all 1,393 lines discounted by more than 20%, 1,348 were loss-making and together they lost 135,376.06. By comparison, the lines discounted 20% or less made 421,773.08 in profit, which is more than the business's entire 286,397.02 profit.

The deep discounts are not spread evenly. Central accounts for 456 of those 933 lines (48.9%), and 19.6% of all Central order lines carry a discount above 40%, against 3.7% in West. By product, Binders make up 613 of the deep-discount lines. Binders are still profitable overall (14.86% margin), so the issue there is the discounted lines inside an otherwise healthy sub-category.

Interpretation: the data shows that heavily discounted lines lose money. It does not show how many of those sales would have happened without the discount, so this should be tested rather than treated as a rule.

4.6 Top 10 Products

The ten best-selling products bring in 244,620.20, which is 10.65% of total sales. Sales are spread across a very wide catalogue (1,841 product names), so no single product dominates. The leader, the **Canon imageCLASS 2200 Advanced Copier**, sold 61,599.82 from only 20 units and made 25,199.93 in profit.



#	Product	Sales	Profit	Units
1	Canon imageCLASS 2200 Advanced Copier	61,599.82	25,199.93	20
2	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27,453.38	7,753.04	31
3	Cisco TelePresence System EX90 Videoconferencing Unit	22,638.48	-1,811.08	6
4	HON 5400 Series Task Chairs for Big and Tall	21,870.58	0.00	39
5	GBC DocuBind TL300 Electric Binding System	19,823.48	2,233.51	37
6	GBC Ibimaster 500 Manual ProClick Binding System	19,024.50	760.98	48
7	Hewlett Packard LaserJet 3310 Copier	18,839.69	6,983.88	38
8	HP Designjet T520 Inkjet Large Format Printer - 24" Color	18,374.90	4,094.98	12
9	GBC DocuBind P400 Electric Binding System	17,965.07	-1,878.17	27
10	High Speed Automatic Electric Letter Opener	17,030.31	-262.00	11

Big sales do not always mean big profit. Of the top 10, 3 products lost money (Cisco TelePresence System EX90, GBC DocuBind P400 and the High Speed Automatic Electric Letter Opener) and the HON 5400 Series Task Chairs finished at break-even. The Canon copier and the HP LaserJet 3310 copier, by contrast, were both clearly profitable.

4.7 Customers and Segments

Consumer customers generate 50.56% of sales (1,161,401.34), **Corporate** 30.74% and **Home Office** 18.70%. Margins are fairly close, from 11.55% for Consumer to 14.03% for Home Office. Revenue is not dependent on a few big accounts: the ten largest customers together account for only 6.7% of sales, and 98.5% of the 793 customers ordered more than once, with an average of 6.3 orders each.

5. Business Recommendations

These recommendations come from the patterns in the workbook. They are actions worth testing, not claims that the data has proved the cause.

1. Put guardrails on discounting

Discounts above 40% lost money on every one of the 933 lines in this data, and discounts above 20% lost money on 1,348 of 1,393. Consider an approval step for discounts above 20% and a firm stop above 40% except for planned clearance. Test the change on a small set of products first so you can see what happens to volume.

2. Fix Central's profit before chasing more Central sales

Central's margin is 7.92% against 14.94% in West, and its average discount is more than double West's. Compare Central's discount rules, product mix and customer mix with West before deciding whether this is a pricing, mix or approval-process problem.

3. Review pricing and cost on Tables, Bookcases and Machines

Tables lose 17,725.48 and Machines barely break even on 189,238.63 of sales. Check supplier cost, shipping and discount levels for these lines. If they exist mainly to attract customers, that is a fair choice, but it should be a deliberate one with a known cost.

4. Judge Furniture on profit, not just sales

Furniture is 32.30% of sales but only 6.4% of profit. Before putting more marketing or stock behind it, fix the loss-making sub-categories. Technology and Office Supplies are where the margin is today, so protect them.

5. Plan stock, staffing and campaigns around the seasonal pattern

September, November and December bring 43.1% of sales, while February is the quietest month. Compare these months with the promotion calendar, product launches and stock levels. If the pattern holds up, build inventory early for the peak and use the quiet months for planning or targeted offers.

6. Work on order size, not only order count

Orders grew 74.6% between 2011 and 2014 while average order value fell 13.2%. Bundles, minimum-spend offers and cross-selling between categories are worth testing, especially on the many small Office Supplies orders.

7. Take a closer look at top sellers that do not earn profit

The Cisco TelePresence System EX90, GBC DocuBind P400 and the High Speed Automatic Electric Letter Opener are all top-10 products by sales but lost money. Review their discounts and costs before promoting or stocking more of them, and see what the profitable Canon and HP copiers are doing differently.

8. Find the reasons before making big moves

The workbook shows what happened, not why. Before dropping a product, changing a price or moving budget between regions, look at the supporting evidence: quantities, discounts, order counts, segments and operational data such as stock and marketing activity.

6. Limitations

- The data covers January 2011 to December 2014 and may not reflect today's market.
- Profit here is the Profit column in the workbook. Overheads, marketing spend and other business costs are not part of it.
- Monthly, regional and discount patterns show association, not proven cause. Information on campaigns, stock and costs is not included.
- The discount analysis works line by line. It cannot show how many sales happened only because of a discount.
- Products are ranked by product name, as in the dashboard. The data has 1,862 product IDs but 1,841 product names, so a few names cover more than one ID.
- Category order counts overlap because one order can include several categories.
- The workbook does not state its original source, so the data should be labelled clearly (for example as a public sample dataset) wherever the project is shared.

7. Conclusion

The business recorded 2,297,200.86 in sales and 286,397.02 in profit across 5,009 orders between 2011 and 2014, and sales grew strongly in 2013 and 2014. West and East lead on sales, Technology is the largest category, and the end of the year is by far the busiest season. The bigger story is profit. Heavy discounting, concentrated in Central and in a handful of sub-categories such as Binders, Machines and Tables, is where a large share of the margin is being given away, while Furniture grows sales without adding much profit. Tightening discount control and fixing the weakest product lines look like the most promising next steps, and each should be tested against order volume, profit and other evidence before it becomes a business decision.

End of report